

–Supplemental Material–

Prune but Preserve: Decoupling Redundancy Suppression and Semantic Preservation for Efficient Time Series Clustering

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I. BASELINE DESCRIPTIONS

The baseline set is organized around the empirical question studied in the main paper, namely at which stage redundancy should be reduced for unsupervised MTS clustering. It covers five streams: full-sequence representation learning, input-side token exclusion, partial-observation clustering, structural sub-sequence indexing, and foundation-style time series modeling. This design separates the benefit of source-level patch admission from gains brought by contrastive objectives, spectral modeling, graph structure, incomplete-view learning, or general temporal pretraining. For embedding-output baselines, k -means gives final assignments with the label-provided cluster number k , following the same unsupervised protocol as the main paper.

FCACC [1] is a fuzzy cluster-aware contrastive clustering method for time series. It constructs augmented views and introduces cluster-aware hard negatives to reduce the gap between instance contrast and cluster assignment. FCACC represents a clustering-oriented full-sequence baseline, testing whether cluster-aware contrast can reduce background influence without source-level patch admission. The official implementation is available at [FCACC](#).

TFMCC [2] performs time-frequency augmented multi-level contrastive clustering. Temporal and spectral augmentations enhance cluster-relevant patterns at different representation levels, but the complete sequence still enters the encoding pipeline. TFMCC tests whether temporal-spectral contrast can handle redundancy as effectively as pruning before encoder-side interaction. The official implementation is available at [TFMCC](#).

GTM [3] learns general time series representations through frequency-domain attention and shared temporal modeling. As GTM is designed for transferable representation rather than clustering-specific reduction, k -means is applied to its embeddings for final assignment. The comparison examines whether broad temporal pretraining can replace explicit patch

admission under redundancy-dominated MTS records. The official implementation is available at [GTM](#).

EMTC [4] is the closest input-side exclusion baseline for MTS clustering. It suppresses redundancy through importance-aware variate-wise masking and enriches retained content with multi-endogenous views. Its comparison with SIGMA directly tests whether confidence-calibrated admission, routed semantic restoration, and prototype-guided refinement improve clustering after sparse token admission. The official implementation is available at [EMTC](#).

FEI [5] introduces a frequency-domain information bottleneck by masking frequency components and predicting the corresponding embeddings. This design encourages informative spectral structure without manually defined positive and negative pairs. FEI serves as a spectral-reduction baseline, assessing whether patch-level admission still helps beyond frequency-oriented representation learning. The official implementation is available at [FEI](#).

MVCIMTS [6] is a contrastive multi-view clustering method for incomplete MTS. It treats variates as views and learns complementary information through multi-view contrast. Since its partial observation comes from variables instead of temporal patches, MVCIMTS tests whether variable-wise incompleteness brings the same clustering benefit as temporal admission followed by routed restoration. The official implementation is available at [MVCIMTS](#).

k -Graph [7] converts subsequences into graph representations and clusters graph-derived features. By building graphs across multiple subsequence lengths, k -Graph captures recurring local patterns and structural similarity. It provides a data-centric baseline for datasets whose clusters depend on shape correspondence or repeated subsequence structure. The official implementation is available at [\$k\$ -Graph](#).

TimesURL [8] is a universal self-supervised representation method for time series. It combines contrastive learning with temporal constraints to obtain general embeddings. In the prune-and-preserve comparison, TimesURL represents a

full-sequence self-supervised baseline because redundancy is not controlled through an admissible patch set. The official implementation is available at [TimesURL](#).

UNITS [9] is a unified multi-task time series model. It represents task information as tokens in a shared modeling framework and learns transferable representations across domains, temporal scales, and sampling patterns. UNITS provides a foundation-style baseline for testing whether task-unified pre-training is sufficient for redundancy-dominated MTS clustering. Its comparison with SIGMA highlights the role of source-level admission and routed semantic preservation in cluster-oriented representation learning. The official implementation is available at [UNITS](#).

II. IMPLEMENTATION DETAILS

All experiments follow the unsupervised clustering protocol in the main paper. For each UEA dataset, training and test splits are concatenated before clustering, and labels are accessed only for metric computation. Each experiment is repeated with five random seeds, and the mean and standard deviation are reported. This protocol keeps the supplemental material consistent with the main-paper evaluation and makes the supplementary tables comparable to the representative results in Section IV of the full paper.

SIGMA uses Confidence-Calibrated Masked Admission as the default admission strategy. In the masking ablation, source-level admission is removed while all other settings remain unchanged, isolating the effect of controlling patch participation before encoder-side feature interaction. Unless otherwise specified, no input perturbation is added, and Gaussian noise is introduced only in the robustness study.

For patch-level admission, the patch length is set to $\ell_p = 4$. Given $m = \lceil T/\ell_p \rceil$ patches, the retained ratio is $r_a = 0.5$, and the retained patch number is $m_a = \lceil r_a m \rceil$. Thus, the encoder processes a compact patch subset after admission. This setting reduces redundant patch participation before feature interaction while keeping enough temporal context for semantic restoration.

For endogenous-view representation learning, the number of views is $n_v = 2$. Each view encoder uses hidden dimension $d_h = 32$, and the fused clustering embedding has dimension $d_z = 64$. The prototype alignment temperature is $\tau_p = 0.1$. These values give SIGMA a compact representation budget while allowing admitted content to be organized into complementary latent views.

SIGMA is trained for $e_{\max} = 200$ epochs with mini-batch size $b = 64$. Adam is adopted with learning rate $\alpha = 0.001$. The loss weights are $\lambda_{\text{rec}} = 1.0$, $\lambda_{\text{orth}} = 0.1$, and $\lambda_{\text{pa}} = 1.0$, corresponding to reconstruction, inter-view orthogonality, and prototype alignment, respectively. Reconstruction anchors temporal content after admission, orthogonality discourages view homogenization, and prototype alignment sharpens the fused cluster space. Under this configuration, admission control, semantic restoration, and cluster refinement keep distinct roles within the prune-and-preserve pipeline.

III. SUPPLEMENTARY RESULTS

This section provides the supplementary empirical record behind the main paper. The main paper keeps the evaluation compact through representative datasets and condensed diagnostics. The supplemental material reports complete benchmark tables, Bonferroni-Dunn rank tests, full module and route ablations, additional training-loss curves, mask-rate scalability, and the robustness and compactness diagnostics in Fig. 4 under the same unsupervised clustering protocol defined in Section II. These results do not introduce another evaluation setting. They make the main claims traceable at the dataset, component, optimization, statistical, and resource levels.

The organization follows the role assignment in SIGMA. Complete clustering results and rank tests examine whether source-level admission and routed representation learning remain competitive across heterogeneous MTS collections. Ablations verify the contributions of admission, endogenous-view organization, semantic restoration, and prototype-guided refinement. Loss curves inspect whether staged training stays controlled after prototype guidance is introduced. Mask-rate scalability checks whether runtime reduction comes from fewer admitted patches before encoder-side interaction. Robustness and compactness diagnostics further test whether the retained representation remains usable under noise perturbation and reduced embedding dimensions. Together, these supplementary results connect clustering quality, statistical reliability, mechanism attribution, training stability, and computation-aware temporal data organization.

A. Complete Clustering Results on 30 UEA Datasets

The main paper answers Q1 with representative per-dataset results and a Bonferroni-Dunn rank test. This supplemental material subsection provides the complete dataset-level record. TABLE I and TABLE II report ACC, F1, NMI, and ARI on all 30 UEA datasets, covering both the datasets discussed in the main paper and the remaining benchmarks omitted due to page limits. The supplemental material therefore supports Q1 with the full benchmark evidence under the same unsupervised protocol.

The complete tables show that SIGMA’s advantage does not come from a few favorable cases. On datasets with clear cluster-revealing patterns, such as ArticulatoryWordRecognition, BasicMotions, JapaneseVowels, PenDigits, and SelfRegulationSCP1, SIGMA obtains leading or highly competitive scores across multiple metrics. These datasets are important for the prune-and-preserve claim, because sparse class-related changes can be diluted when complete sequences enter feature interaction. By reducing background-dominated patch participation before encoding and organizing retained content through routed objectives, SIGMA keeps compact representations informative for clustering. JapaneseVowels and PenDigits are typical cases where reduced input participation still yields a discriminative grouping space.

The full benchmark also reflects the diversity of unsupervised MTS clustering. Several datasets favor specialized baselines when cluster similarity depends more on shape matching,

TABLE I
COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART I. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND. OOM DENOTES OUT-OF-MEMORY RUNS.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	k-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
ArticulatoryWordRecognition	ACC	0.7329 ± 0.0037	0.3555 ± 0.0380	0.3547 ± 0.0185	0.5030 ± 0.0253	0.4710 ± 0.0731	0.0497 ± 0.0124	0.2838 ± 0.0159	0.6929 ± 0.0516	0.2948 ± 0.0200	0.7603 ± 0.0232
	F1	0.8039 ± 0.0055	0.3526 ± 0.0348	0.3560 ± 0.0214	0.3994 ± 0.0241	0.4974 ± 0.0815	0.0109 ± 0.0096	0.2774 ± 0.0169	0.7425 ± 0.0413	0.3114 ± 0.0243	0.7929 ± 0.0392
	NMI	0.8408 ± 0.0064	0.4708 ± 0.0311	0.5070 ± 0.0272	0.7012 ± 0.0159	0.6185 ± 0.0744	0.0314 ± 0.0428	0.3948 ± 0.0099	0.8215 ± 0.0145	0.4280 ± 0.0198	0.8276 ± 0.0165
	ARI	0.6641 ± 0.0045	0.1833 ± 0.0296	0.1615 ± 0.0271	0.4466 ± 0.0210	0.3352 ± 0.0843	0.0019 ± 0.0031	0.1224 ± 0.0091	0.6214 ± 0.0340	0.1395 ± 0.0188	0.6466 ± 0.0257
AtrialFibrillation	ACC	0.4333 ± 0.0000	0.4667 ± 0.0422	0.6133 ± 0.0499	0.4333 ± 0.0558	0.3867 ± 0.0163	0.4095 ± 0.0440	0.4400 ± 0.0389	0.4600 ± 0.0490	0.4500 ± 0.0167	0.5000 ± 0.0365
	F1	0.4303 ± 0.0028	0.4593 ± 0.0393	0.5828 ± 0.0633	0.4041 ± 0.0191	0.3543 ± 0.0194	0.3491 ± 0.0555	0.4295 ± 0.0406	0.4169 ± 0.0278	0.4343 ± 0.0089	0.4956 ± 0.0778
	NMI	0.1005 ± 0.0040	0.0726 ± 0.0277	0.2883 ± 0.0758	0.0544 ± 0.0351	0.0353 ± 0.0237	0.1445 ± 0.0821	0.1143 ± 0.0748	0.0858 ± 0.0339	0.0643 ± 0.0072	0.1251 ± 0.0390
	ARI	-0.0056 ± 0.0011	0.0051 ± 0.0258	0.1248 ± 0.0788	0.0064 ± 0.0282	-0.0360 ± 0.0089	0.1058 ± 0.0600	0.0039 ± 0.0462	0.0136 ± 0.0342	-0.0125 ± 0.0012	0.0582 ± 0.0393
BasicMotions	ACC	0.6675 ± 0.0100	0.7300 ± 0.1086	0.7950 ± 0.0292	0.9850 ± 0.0122	0.6775 ± 0.0443	0.2975 ± 0.0827	0.8875 ± 0.1321	0.7700 ± 0.1582	0.5437 ± 0.0812	0.9875 ± 0.0079
	F1	0.8228 ± 0.0098	0.7311 ± 0.1060	0.7832 ± 0.0275	0.9701 ± 0.0235	0.7351 ± 0.0723	0.1517 ± 0.0794	0.8792 ± 0.1500	0.8627 ± 0.0835	0.5705 ± 0.0412	0.9875 ± 0.0079
	NMI	0.8048 ± 0.0020	0.5569 ± 0.1633	0.6949 ± 0.0711	0.9627 ± 0.0248	0.6792 ± 0.0317	0.1007 ± 0.1775	0.8626 ± 0.0575	0.8487 ± 0.0634	0.3898 ± 0.1107	0.9665 ± 0.0191
	ARI	0.6295 ± 0.0045	0.4526 ± 0.1902	0.6085 ± 0.0606	0.9606 ± 0.0311	0.5317 ± 0.0351	0.0690 ± 0.1381	0.8174 ± 0.1218	0.7349 ± 0.1448	0.2989 ± 0.1276	0.9666 ± 0.0208
CharacterTrajectories	ACC	0.7981 ± 0.0012	0.0651 ± 0.0000	0.3000 ± 0.0261	0.3677 ± 0.0448	0.6040 ± 0.0096	0.0866 ± 0.0318	0.4500 ± 0.0167	0.0651 ± 0.0000	0.4533 ± 0.0267	0.7668 ± 0.0529
	F1	0.8583 ± 0.0012	0.0601 ± 0.0000	0.2886 ± 0.0434	0.3405 ± 0.0281	0.6289 ± 0.0099	0.0196 ± 0.0166	0.4350 ± 0.0149	0.1222 ± 0.0000	0.1703 ± 0.0373	0.8019 ± 0.0378
	NMI	0.8726 ± 0.0013	0.0000 ± 0.0000	0.3633 ± 0.0432	0.5756 ± 0.0418	0.7095 ± 0.0041	0.0682 ± 0.1107	0.5347 ± 0.0121	0.0000 ± 0.0000	0.1776 ± 0.0668	0.8261 ± 0.0315
	ARI	0.7705 ± 0.0012	0.0000 ± 0.0000	0.1756 ± 0.0296	0.1913 ± 0.0474	0.5137 ± 0.0138	0.0181 ± 0.0345	0.3211 ± 0.0193	0.0000 ± 0.0000	0.0532 ± 0.0279	0.6933 ± 0.0563
Cricket	ACC	0.9407 ± 0.0288	0.4533 ± 0.0264	0.5528 ± 0.0204	0.6922 ± 0.0664	0.5744 ± 0.0834	0.1167 ± 0.0350	0.4589 ± 0.0218	0.6722 ± 0.0813	0.4922 ± 0.0990	0.8489 ± 0.0787
	F1	0.9599 ± 0.0024	0.4393 ± 0.0376	0.5635 ± 0.0383	0.6319 ± 0.0703	0.6027 ± 0.0590	0.0417 ± 0.0250	0.4545 ± 0.0225	0.7562 ± 0.0557	0.5414 ± 0.0992	0.8838 ± 0.0523
	NMI	0.9509 ± 0.0020	0.5189 ± 0.0101	0.6181 ± 0.0286	0.7858 ± 0.0464	0.6794 ± 0.0579	0.0817 ± 0.0964	0.4752 ± 0.0228	0.7958 ± 0.0391	0.5947 ± 0.0841	0.8764 ± 0.0445
	ARI	0.9051 ± 0.0158	0.5284 ± 0.0194	0.2971 ± 0.0365	0.5842 ± 0.0827	0.4468 ± 0.0872	0.0171 ± 0.0299	0.2533 ± 0.0250	0.6028 ± 0.0783	0.3359 ± 0.1072	0.7672 ± 0.0967
DuckDuckGeese	ACC	0.3067 ± 0.0125	0.4140 ± 0.0162	0.4240 ± 0.0408	0.4720 ± 0.0172	0.3220 ± 0.0117	0.2620 ± 0.0519	0.3280 ± 0.0160	0.3220 ± 0.0147	0.2960 ± 0.0136	0.4020 ± 0.0325
	F1	0.2611 ± 0.0200	0.4042 ± 0.0246	0.4003 ± 0.0494	0.3394 ± 0.0193	0.3013 ± 0.0347	0.1650 ± 0.0575	0.3017 ± 0.0107	0.2641 ± 0.0182	0.2869 ± 0.0075	0.3819 ± 0.0291
	NMI	0.0800 ± 0.0179	0.2483 ± 0.0205	0.2082 ± 0.0380	0.2791 ± 0.0267	0.1030 ± 0.0090	0.0335 ± 0.0301	0.0958 ± 0.0101	0.1732 ± 0.0351	0.0682 ± 0.0175	0.1718 ± 0.0577
	ARI	0.0227 ± 0.0109	0.1394 ± 0.0210	0.0729 ± 0.0331	0.1780 ± 0.0248	0.0369 ± 0.0091	0.0073 ± 0.0170	0.0225 ± 0.0087	0.0375 ± 0.0110	0.0064 ± 0.0105	0.1000 ± 0.0580
EigenWorms	ACC	OOM	0.3876 ± 0.0072	OOM	OOM	0.4185 ± 0.0361	0.4324 ± 0.0024	0.3135 ± 0.0236	0.3660 ± 0.0510	0.4525 ± 0.0366	0.4332 ± 0.0402
	F1	OOM	0.2868 ± 0.0068	OOM	OOM	0.3899 ± 0.0605	0.1545 ± 0.0173	0.2920 ± 0.0180	0.3035 ± 0.0806	0.4410 ± 0.0601	0.4106 ± 0.0483
	NMI	OOM	0.0683 ± 0.0035	OOM	OOM	0.1826 ± 0.0152	0.0300 ± 0.0094	0.0848 ± 0.0181	0.0697 ± 0.0572	0.2872 ± 0.0919	0.1885 ± 0.0300
	ARI	OOM	0.0626 ± 0.0052	OOM	OOM	0.1453 ± 0.0227	0.0098 ± 0.0046	0.0400 ± 0.0160	0.0515 ± 0.0279	0.1694 ± 0.0486	0.1620 ± 0.0458
Epilepsy	ACC	0.4279 ± 0.0297	0.3869 ± 0.0404	0.5203 ± 0.0244	0.4953 ± 0.0477	0.6364 ± 0.0808	0.2953 ± 0.0324	0.6167 ± 0.0316	0.5156 ± 0.0189	0.4982 ± 0.0646	0.4756 ± 0.0321
	F1	0.4295 ± 0.0302	0.3710 ± 0.0430	0.5136 ± 0.0265	0.4147 ± 0.0552	0.6200 ± 0.0821	0.1882 ± 0.0235	0.6200 ± 0.0260	0.4979 ± 0.0080	0.5086 ± 0.0930	0.4724 ± 0.0357
	NMI	0.1397 ± 0.0347	0.1015 ± 0.0345	0.2476 ± 0.0419	0.3029 ± 0.0794	0.4079 ± 0.0773	0.0205 ± 0.0340	0.4013 ± 0.0415	0.3363 ± 0.0795	0.3126 ± 0.0851	0.2652 ± 0.0289
	ARI	0.0960 ± 0.0211	0.0814 ± 0.0434	0.1803 ± 0.0389	0.1992 ± 0.0638	0.3829 ± 0.0804	0.0076 ± 0.0184	0.3275 ± 0.0526	0.2288 ± 0.0542	0.2320 ± 0.0733	0.1889 ± 0.0197
ERing	ACC	0.6327 ± 0.0360	0.4180 ± 0.0585	0.6704 ± 0.0486	0.8607 ± 0.0218	0.7120 ± 0.0918	0.3667 ± 0.1560	0.5380 ± 0.0421	0.7433 ± 0.1161	0.4087 ± 0.0639	0.7980 ± 0.0749
	F1	0.6306 ± 0.0393	0.4117 ± 0.0586	0.6741 ± 0.0496	0.7564 ± 0.0332	0.7508 ± 0.0811	0.2682 ± 0.1646	0.5469 ± 0.0403	0.7862 ± 0.0839	0.4159 ± 0.0698	0.8003 ± 0.0701
	NMI	0.5660 ± 0.0104	0.2357 ± 0.0481	0.5487 ± 0.0473	0.7821 ± 0.0321	0.6562 ± 0.0772	0.3054 ± 0.2115	0.3843 ± 0.0143	0.7407 ± 0.0313	0.2528 ± 0.0802	0.6872 ± 0.0892
	ARI	0.4231 ± 0.0204	0.1602 ± 0.0466	0.4382 ± 0.0499	0.7051 ± 0.0415	0.5669 ± 0.0959	0.1897 ± 0.1599	0.2816 ± 0.0319	0.6467 ± 0.0757	0.1578 ± 0.0647	0.6168 ± 0.1270
EthanolConcentration	ACC	0.2992 ± 0.0025	0.2786 ± 0.0051	0.3605 ± 0.0224	0.2634 ± 0.0096	0.2901 ± 0.0017	0.2962 ± 0.0172	0.2920 ± 0.0100	0.2962 ± 0.0044	0.2691 ± 0.0034	0.2969 ± 0.0153
	F1	0.2759 ± 0.0027	0.2529 ± 0.0093	0.3458 ± 0.0279	0.4681 ± 0.0189	0.2744 ± 0.0087	0.2442 ± 0.0306	0.2882 ± 0.0110	0.2747 ± 0.0067	0.2090 ± 0.0059	0.2732 ± 0.0102
	NMI	0.0149 ± 0.0009	0.0047 ± 0.0010	0.0574 ± 0.0189	0.0124 ± 0.0098	0.0116 ± 0.0018	0.0176 ± 0.0112	0.0074 ± 0.0029	0.0199 ± 0.0025	0.0025 ± 0.0015	0.0190 ± 0.0157
	ARI	0.0012 ± 0.0003	-0.0013 ± 0.0010	0.0307 ± 0.0121	0.0003 ± 0.0007	0.0032 ± 0.0005	0.0072 ± 0.0073	0.0008 ± 0.0024	0.0039 ± 0.0009	-0.0014 ± 0.0010	0.0097 ± 0.0119
FaceDetection	ACC	0.5130 ± 0.0030	OOM	0.5146 ± 0.0047	0.5051 ± 0.0029	0.5077 ± 0.0031	OOM	0.5038 ± 0.0018	0.5001 ± 0.0000	0.5054 ± 0.0040	0.5198 ± 0.0044
	F1	0.5125 ± 0.0028	OOM	0.4808 ± 0.0369	0.6113 ± 0.0585	0.5051 ± 0.0039	OOM	0.5028 ± 0.0023	0.3336 ± 0.0000	0.5030 ± 0.0060	0.5192 ± 0.0046
	NMI	0.0005 ± 0.0002	OOM	0.0010 ± 0.0004	0.0002 ± 0.0002	0.0002 ± 0.0001	OOM	0.0001 ± 0.0000	0.0027 ± 0.0000	0.0001 ± 0.0002	0.0012 ± 0.0005
	ARI	0.0006 ± 0.0003	OOM	0.0007 ± 0.0006	0.0001 ± 0.0001	0.0002 ± 0.0002	OOM	-0.0000 ± 0.0000	0.0000 ± 0.0000	0.0001 ± 0.0003	0.0015 ± 0.0006
FingerMovements	ACC	0.5250 ± 0.0052	0.5183 ± 0.0085	0.5680 ± 0.0160	0.5072 ± 0.0055	0.5163 ± 0.0127	0.5260 ± 0.0041	0.5264 ± 0.0026	0.5192 ± 0.0082	0.5130 ± 0.0054	0.5221 ± 0.0129
	F1	0.4951 ± 0.0111	0.4982 ± 0.0212	0.5349 ± 0.0218	0.6004 ± 0.0599	0.5051 ± 0.0184	0.5230 ± 0.0023	0.4997 ± 0.0023	0.5161 ± 0.0058	0.5124 ± 0.0051	0.5183 ± 0.0074
	NMI	0.0027 ± 0.0009	0.0014 ± 0.0009	0.0210 ± 0.0132	0.0023 ± 0.0035	0.0013 ± 0.0016	0.0021 ± 0.0007	0.0028 ± 0.0005	0.0013 ± 0.0009	0.0006 ± 0.0003	0.0022 ± 0.0026
	ARI	0.0008 ± 0.0009	-0.0004 ± 0.0009	0.0121 ± 0.0087	-0.0010 ± 0.0008	-0.0005 ± 0.0021	0.0004 ± 0.0009	0.0009 ± 0.0005	-0.0006 ± 0.0013	-0.0016 ± 0.0004	0.0003 ± 0.0031
HandMovementDirection	ACC	0.2829 ± 0.0063	0.2957 ± 0.0087	0.4405 ± 0.0359	0.3248 ± 0.0164	0.3017 ± 0.0088	0.3017 ± 0.0123	0.3026 ± 0.0099	0.3120 ± 0.0097	0.3743 ± 0.0375	0.3179 ± 0.0100
	F1	0.2108 ± 0.0397	0.2907 ± 0.0061	0.4070 ± 0.0226	0.3328 ± 0.0189	0.2867 ± 0.0112	0.2996 ± 0.0128	0.3080 ± 0.0099	0.3080 ± 0.0099	0.3667 ± 0.0383	0.3145 ± 0.0104
	NMI	0.0168 ± 0.0104	0.0156 ± 0.0061	0.0910 ± 0.0245	0.0286 ± 0.0110	0.0184 ± 0.0049	0.0206 ± 0.0103	0.0144 ± 0.0047	0.0208 ± 0.0045	0.0818 ± 0.0394	0.0208 ± 0.0050
	ARI	-0.0047 ± 0.0038	0.0015 ± 0.0062	0.0475 ± 0.0166	0.0109 ± 0.0104	0.0001 ± 0.0045	0.0007 ± 0.0032	0.0003 ± 0.0032	0.0046 ± 0.0035	0.0488 ± 0.0282	0.0052 ± 0.0047
Handwriting	ACC	0.2433 ± 0.0041	0.1264 ± 0.0027	0.1536 ± 0.0097	0.1090 ± 0.0209	0.1778 ± 0.0097	0.0878 ± 0.0160	0.1708 ± 0.0101	0.1488 ± 0.0067	0.2940 ± 0.0068	0.2530 ± 0.0126
	F1	0.2418 ± 0.0080	0.1260 ± 0.0020	0.1501 ± 0.0114	0.1206 ± 0.0171	0.1741 ± 0.0071	0.0352 ± 0.0127	0.1679 ± 0.0098	0.1484 ± 0.0078	0.2881 ± 0.0075	0.2591 ± 0.0132
	NMI	0.3723 ± 0.0061	0.1523 ± 0.0052	0.1934 ± 0.0070	0.1228 ± 0.0510	0.2497 ± 0.0235	0.0637 ± 0.0274	0.2260 ± 0.0097	0.2009 ± 0.0035	0.0101 ± 0.0049	0.3652 ± 0.0139
	ARI	0.1106 ± 0.0035	0.0125 ± 0.0016	0.0254 ± 0.0036	0.0195 ± 0.0090	0.0494 ± 0.0107	0.0084 ± 0.0048	0.0429 ± 0.0045	0.0282 ± 0.0012	-0.0047 ± 0.0030	0.1134 ± 0.0113

spectral structure, or weak label-to-signal correspondence. For example, FCACC, TFMCC, EMTC, FEI, and k -Graph obtain the best result on some individual datasets. Thus, the key observation is not dominance on every dataset, but stable average behavior across different sample sizes, sequence lengths, variable dimensions, and label-provided cluster numbers k . Fig. 1 provides the corresponding Bonferroni-Dunn test. Average ranks are computed over 30 datasets and four metrics, and lower ranks indicate stronger overall clustering quality. The critical-difference bars show which rank gaps are statistically distinguishable at $\alpha = 0.1$ and $\alpha = 0.05$, giving a statistical view of the cross-dataset consistency reported in the tables.

B. Complete Module-Level and Route-Level Ablations

The main paper reports averaged ablations for Q2, and this supplemental material subsection expands the same analysis to all 30 UEA datasets. All variants follow the protocol in Section II. The module-level ablations remove CCMA and MEVG separately or jointly, testing how patch participation

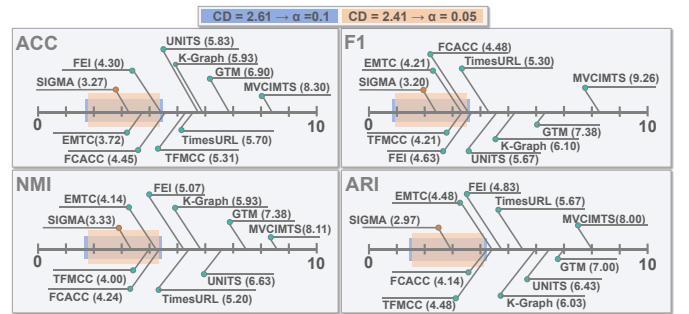


TABLE II
COMPLETE CLUSTERING RESULTS ON 30 UEA DATASETS EVALUATED BY ACC, F1, NMI, AND ARI, PART II. THE **BEST** AND **SECOND BEST** RESULTS ARE HIGHLIGHTED BY CELL BACKGROUND.

Datasets	Metric	FCACC (PR'26)	GTM (ICLR'26)	EMTC (AAAI'26)	TFMCC (AAAI'26)	FEI (AAAI'25)	MVCIMTS (INFFUS'25)	k-Graph (TKDE'25)	TimesURL (AAAI'24)	UNITS (NeurIPS'24)	SIGMA (ours)
Heartbeat	ACC	0.5844 ± 0.0079	0.5340 ± 0.0292	0.7180 ± 0.0020	0.7076 ± 0.0476	0.7090 ± 0.0000	0.6758 ± 0.0497	0.7296 ± 0.0025	0.7149 ± 0.0106	0.6244 ± 0.0238	0.6817 ± 0.0546
	F1	0.5688 ± 0.0055	0.4974 ± 0.0193	0.7553 ± 0.1598	0.6934 ± 0.0669	0.8298 ± 0.0000	0.4592 ± 0.0483	0.5461 ± 0.0071	0.7922 ± 0.0760	0.5347 ± 0.0192	0.5944 ± 0.0367
	NMI	0.0389 ± 0.0016	0.0012 ± 0.0015	0.0106 ± 0.0023	0.0465 ± 0.0073	0.0204 ± 0.0000	0.0016 ± 0.0020	0.0344 ± 0.0041	0.0300 ± 0.0207	0.0070 ± 0.0077	0.0382 ± 0.0220
	ARI	0.0258 ± 0.0062	0.0030 ± 0.0062	-0.0029 ± 0.0061	0.0966 ± 0.0306	-0.0144 ± 0.0000	0.0018 ± 0.0101	0.0875 ± 0.0071	0.0220 ± 0.0714	0.0243 ± 0.0111	0.0981 ± 0.0485
InsectWingbeat	ACC	0.1651 ± 0.0020	0.1003 ± 0.0000	0.1251 ± 0.0049	0.1099 ± 0.0024	0.1533 ± 0.0004	OOM	0.1451 ± 0.0010	0.1000 ± 0.0000	0.0872 ± 0.0071	0.1336 ± 0.0017
	F1	0.1623 ± 0.0020	0.0189 ± 0.0001	0.0945 ± 0.0222	0.0962 ± 0.0074	0.1062 ± 0.0051	OOM	0.1374 ± 0.0049	0.1818 ± 0.0000	0.0812 ± 0.0055	0.1209 ± 0.0043
	NMI	0.0428 ± 0.0011	0.0010 ± 0.0001	0.0090 ± 0.0015	0.1883 ± 0.0060	0.0339 ± 0.0008	OOM	0.0210 ± 0.0008	0.0000 ± 0.0000	0.0032 ± 0.0061	0.0146 ± 0.0015
	ARI	0.0216 ± 0.0005	0.0000 ± 0.0000	0.0027 ± 0.0013	0.0377 ± 0.0055	0.0130 ± 0.0007	OOM	0.0087 ± 0.0002	0.0000 ± 0.0000	0.0043 ± 0.0000	0.0059 ± 0.0006
JapaneseVowels	ACC	0.6286 ± 0.0530	0.1844 ± 0.0000	0.3203 ± 0.0273	0.4766 ± 0.0639	0.5198 ± 0.0223	0.1844 ± 0.0000	0.3369 ± 0.0199	0.1844 ± 0.0000	0.2619 ± 0.0271	0.8525 ± 0.0431
	F1	0.6617 ± 0.0303	0.0346 ± 0.0000	0.3325 ± 0.0365	0.4274 ± 0.0355	0.5551 ± 0.0053	0.0346 ± 0.0000	0.3069 ± 0.0236	0.3113 ± 0.0000	0.2755 ± 0.0282	0.8494 ± 0.0383
	NMI	0.5950 ± 0.0134	0.0000 ± 0.0000	0.2410 ± 0.0418	0.5250 ± 0.0397	0.4727 ± 0.0084	0.0000 ± 0.0000	0.2820 ± 0.0112	0.0000 ± 0.0000	0.1643 ± 0.0209	0.8113 ± 0.0180
	ARI	0.4771 ± 0.0389	0.0000 ± 0.0000	0.1269 ± 0.0412	0.3167 ± 0.0466	0.3049 ± 0.0281	0.0000 ± 0.0000	0.1511 ± 0.0210	0.0000 ± 0.0000	0.0648 ± 0.0171	0.7578 ± 0.0423
Libras	ACC	0.3822 ± 0.0052	0.3894 ± 0.0321	0.4011 ± 0.0315	0.3544 ± 0.0099	0.4317 ± 0.0256	0.1456 ± 0.0416	0.3844 ± 0.0208	0.2700 ± 0.0148	0.2839 ± 0.0116	0.3711 ± 0.0125
	F1	0.3931 ± 0.0193	0.3772 ± 0.0369	0.4028 ± 0.0406	0.2626 ± 0.0226	0.4438 ± 0.0298	0.0627 ± 0.0302	0.3700 ± 0.0242	0.2767 ± 0.0166	0.2853 ± 0.0157	0.3655 ± 0.0158
	NMI	0.5311 ± 0.0095	0.4697 ± 0.0327	0.4890 ± 0.0322	0.4698 ± 0.0245	0.5056 ± 0.0189	0.1311 ± 0.0783	0.5126 ± 0.0062	0.3101 ± 0.0185	0.3562 ± 0.0152	0.4086 ± 0.0259
	ARI	0.2484 ± 0.0082	0.2143 ± 0.0276	0.1881 ± 0.0295	0.2019 ± 0.0228	0.2423 ± 0.0142	0.0370 ± 0.0255	0.2532 ± 0.0127	0.0901 ± 0.0166	0.1246 ± 0.0146	0.2809 ± 0.0196
LSST	ACC	0.3434 ± 0.0022	0.1224 ± 0.0038	0.3672 ± 0.0074	0.2491 ± 0.0169	0.3327 ± 0.0037	0.3227 ± 0.0092	0.1578 ± 0.0113	0.3293 ± 0.0025	0.1805 ± 0.0098	0.2019 ± 0.0029
	F1	0.1918 ± 0.0134	0.0984 ± 0.0072	0.2055 ± 0.0297	0.2114 ± 0.0148	0.1689 ± 0.0141	0.0521 ± 0.0222	0.1149 ± 0.0053	0.2387 ± 0.0157	0.1376 ± 0.0084	0.1840 ± 0.0147
	NMI	0.1188 ± 0.0020	0.0311 ± 0.0056	0.1959 ± 0.0261	0.2066 ± 0.0198	0.1338 ± 0.0040	0.0237 ± 0.0303	0.0764 ± 0.0061	0.1398 ± 0.0038	0.0543 ± 0.0132	0.1396 ± 0.0043
	ARI	0.0453 ± 0.0007	0.0096 ± 0.0022	0.0936 ± 0.0182	0.0634 ± 0.0311	0.0859 ± 0.0106	0.0070 ± 0.0089	0.0222 ± 0.0040	0.0254 ± 0.0061	0.0207 ± 0.0068	0.0549 ± 0.0028
MotorImagery	ACC	0.5016 ± 0.0013	0.5360 ± 0.0147	0.5840 ± 0.0258	0.5048 ± 0.0072	0.5058 ± 0.0035	0.5111 ± 0.0059	0.5159 ± 0.0111	0.5127 ± 0.0048	0.5333 ± 0.0173	0.5508 ± 0.0070
	F1	0.3512 ± 0.0022	0.5084 ± 0.0319	0.5308 ± 0.0377	0.6586 ± 0.0706	0.3794 ± 0.0205	0.4565 ± 0.0416	0.5093 ± 0.0107	0.4401 ± 0.0338	0.5180 ± 0.0271	0.5362 ± 0.0175
	NMI	0.0003 ± 0.0002	0.0055 ± 0.0031	0.0540 ± 0.0291	0.0023 ± 0.0040	0.0011 ± 0.0008	0.0017 ± 0.0016	0.0012 ± 0.0011	0.0012 ± 0.0004	0.0044 ± 0.0042	0.0096 ± 0.0034
	ARI	-0.0002 ± 0.0000	0.0040 ± 0.0035	0.0254 ± 0.0160	-0.0003 ± 0.0004	-0.0003 ± 0.0001	-0.0010 ± 0.0011	-0.0010 ± 0.0014	-0.0006 ± 0.0001	0.0033 ± 0.0059	0.0082 ± 0.0028
NATOPS	ACC	0.6537 ± 0.0975	0.4861 ± 0.0512	0.3978 ± 0.0514	0.5672 ± 0.0859	0.4350 ± 0.0268	0.3267 ± 0.0919	0.3817 ± 0.0194	0.3522 ± 0.0811	0.2972 ± 0.0249	0.5022 ± 0.0157
	F1	0.6529 ± 0.0968	0.4643 ± 0.0539	0.4342 ± 0.1010	0.5167 ± 0.0816	0.4372 ± 0.0599	0.2057 ± 0.0868	0.3871 ± 0.0168	0.2131 ± 0.0691	0.2951 ± 0.0249	0.4958 ± 0.0103
	NMI	0.6454 ± 0.0805	0.4355 ± 0.0551	0.2990 ± 0.1190	0.5575 ± 0.0724	0.3896 ± 0.0459	0.2876 ± 0.1673	0.1897 ± 0.0207	0.4115 ± 0.1613	0.1097 ± 0.0324	0.5159 ± 0.0230
	ARI	0.5091 ± 0.1162	0.3056 ± 0.0590	0.1759 ± 0.0826	0.4179 ± 0.0983	0.2563 ± 0.0459	0.1603 ± 0.1191	0.1147 ± 0.0214	0.2551 ± 0.1261	0.0549 ± 0.0222	0.3489 ± 0.0284
PEMS-SF	ACC	0.4864 ± 0.0558	0.1968 ± 0.0065	0.4301 ± 0.0842	0.2268 ± 0.0179	0.3582 ± 0.0366	0.2514 ± 0.0897	0.4268 ± 0.0172	0.2377 ± 0.0102	0.4409 ± 0.0466	0.4436 ± 0.0844
	F1	0.5110 ± 0.0447	0.1941 ± 0.0064	0.4202 ± 0.0807	0.2263 ± 0.0204	0.3782 ± 0.0333	0.1185 ± 0.0734	0.4239 ± 0.0214	0.2370 ± 0.0154	0.4683 ± 0.0591	0.4902 ± 0.1198
	NMI	0.4510 ± 0.0416	0.0348 ± 0.0119	0.3687 ± 0.1015	0.1270 ± 0.0168	0.3336 ± 0.0281	0.2715 ± 0.2226	0.4593 ± 0.0196	0.1742 ± 0.0226	0.4167 ± 0.0441	0.4750 ± 0.1104
	ARI	0.2911 ± 0.0478	0.0030 ± 0.0034	0.2134 ± 0.0924	0.0374 ± 0.0156	0.1673 ± 0.0312	0.1333 ± 0.1105	0.2557 ± 0.0121	0.0651 ± 0.0082	0.2471 ± 0.0435	0.2839 ± 0.0947
PenDigits	ACC	0.4246 ± 0.0409	0.4128 ± 0.0672	0.4089 ± 0.0499	0.6548 ± 0.0159	0.3484 ± 0.0354	0.2670 ± 0.0505	0.4078 ± 0.0162	0.3647 ± 0.0150	0.4687 ± 0.0517	0.6841 ± 0.0284
	F1	0.4136 ± 0.0303	0.3971 ± 0.0746	0.4086 ± 0.0578	0.5693 ± 0.0171	0.3548 ± 0.0321	0.1710 ± 0.0688	0.3825 ± 0.0213	0.3747 ± 0.0150	0.4542 ± 0.0512	0.6938 ± 0.0411
	NMI	0.3233 ± 0.0199	0.3405 ± 0.0616	0.3994 ± 0.0617	0.6583 ± 0.0086	0.3059 ± 0.0165	0.2076 ± 0.0559	0.3390 ± 0.0131	0.2611 ± 0.0165	0.4348 ± 0.0630	0.6485 ± 0.0189
	ARI	0.2218 ± 0.0289	0.2151 ± 0.0560	0.2313 ± 0.0515	0.5196 ± 0.0193	0.1503 ± 0.0122	0.1033 ± 0.0425	0.2183 ± 0.0158	0.1528 ± 0.0112	0.2822 ± 0.0658	0.5305 ± 0.0276
PhonemeSpectra	ACC	0.1149 ± 0.0003	0.0633 ± 0.0006	0.0963 ± 0.0032	0.1124 ± 0.0038	0.1283 ± 0.0024	0.0454 ± 0.0059	0.0916 ± 0.0028	0.0871 ± 0.0023	0.0812 ± 0.0071	0.1127 ± 0.0015
	F1	0.1159 ± 0.0004	0.0605 ± 0.0010	0.0944 ± 0.0004	0.1048 ± 0.0067	0.1242 ± 0.0038	0.0099 ± 0.0028	0.0913 ± 0.0033	0.0845 ± 0.0017	0.0786 ± 0.0080	0.1126 ± 0.0017
	NMI	0.1793 ± 0.0023	0.0451 ± 0.0015	0.1386 ± 0.0089	0.2039 ± 0.0025	0.2085 ± 0.0038	0.0366 ± 0.0136	0.1090 ± 0.0036	0.1174 ± 0.0070	0.0932 ± 0.0191	0.1758 ± 0.0026
	ARI	0.0339 ± 0.0007	0.0038 ± 0.0003	0.0166 ± 0.0040	0.0424 ± 0.0048	0.0449 ± 0.0011	0.0046 ± 0.0026	0.0178 ± 0.0015	0.0186 ± 0.0020	0.0143 ± 0.0045	0.0334 ± 0.0008
RocketSports	ACC	0.4172 ± 0.0058	0.3921 ± 0.0150	0.4355 ± 0.0286	0.6205 ± 0.0400	0.4257 ± 0.0251	0.3129 ± 0.0296	0.3551 ± 0.0324	0.5036 ± 0.0475	0.3743 ± 0.0375	0.6007 ± 0.0721
	F1	0.4138 ± 0.0065	0.3898 ± 0.0135	0.4064 ± 0.0374	0.5330 ± 0.0327	0.4067 ± 0.0328	0.1205 ± 0.0508	0.3529 ± 0.0340	0.5273 ± 0.0345	0.3667 ± 0.0383	0.6006 ± 0.0707
	NMI	0.1398 ± 0.0042	0.1024 ± 0.0386	0.1138 ± 0.0264	0.4687 ± 0.0339	0.1250 ± 0.0289	0.0688 ± 0.0371	0.0601 ± 0.0348	0.3279 ± 0.0420	0.0818 ± 0.0394	0.3488 ± 0.0845
	ARI	0.0880 ± 0.0042	0.0590 ± 0.0167	0.0776 ± 0.0307	0.3723 ± 0.0445	0.0924 ± 0.0200	0.0127 ± 0.0191	0.0400 ± 0.0293	0.2394 ± 0.0441	0.0488 ± 0.0282	0.2858 ± 0.0965
SelfRegulationSCP1	ACC	0.5362 ± 0.0031	0.5447 ± 0.0307	0.7126 ± 0.0431	0.5027 ± 0.0000	0.6096 ± 0.0472	0.5537 ± 0.0342	0.5586 ± 0.0087	0.7340 ± 0.0059	0.5811 ± 0.0402	0.7754 ± 0.0128
	F1	0.5081 ± 0.0040	0.5434 ± 0.0324	0.7073 ± 0.0419	0.7065 ± 0.0000	0.6082 ± 0.0487	0.4723 ± 0.0964	0.5584 ± 0.0087	0.7184 ± 0.0076	0.5676 ± 0.0372	0.7745 ± 0.0125
	NMI	0.0059 ± 0.0009	0.0086 ± 0.0093	0.1642 ± 0.0748	0.0000 ± 0.0000	0.0421 ± 0.0290	0.0261 ± 0.0139	0.0102 ± 0.0031	0.2493 ± 0.0044	0.0328 ± 0.0305	0.2382 ± 0.0246
	ARI	0.0039 ± 0.0009	0.0100 ± 0.0128	0.1857 ± 0.0753	0.0000 ± 0.0000	0.0553 ± 0.0393	0.0154 ± 0.0124	0.0123 ± 0.0043	0.2182 ± 0.0111	0.0313 ± 0.0296	0.3028 ± 0.0274
SelfRegulationSCP2	ACC	0.5163 ± 0.0051	0.5258 ± 0.0162	0.5811 ± 0.0143	0.5027 ± 0.0000	0.5521 ± 0.0096	0.5347 ± 0.0210	0.5053 ± 0.0069	0.5432 ± 0.0054	0.5811 ± 0.0402	0.5253 ± 0.0141
	F1	0.4913 ± 0.0063	0.5242 ± 0.0166	0.5657 ± 0.0301	0.7065 ± 0.0000	0.5448 ± 0.0077	0.5302 ± 0.0239	0.5039 ± 0.0073	0.5113 ± 0.0110	0.5676 ± 0.0372	0.5163 ± 0.0084
	NMI	0.0011 ± 0.0006	0.0027 ± 0.0027	0.0234 ± 0.0043	0.0000 ± 0.0000	0.0090 ± 0.0034	0.0049 ± 0.0048	0.0002 ± 0.0004	0.0083 ± 0.0020	0.0328 ± 0.0305	0.0034 ± 0.0044
	ARI	-0.0010 ± 0.0006	0.0111 ± 0.0036	0.0224 ± 0.0083	0.0000 ± 0.0000	0.0088 ± 0.0040	0.0041 ± 0.0065	-0.0023 ± 0.0005	0.0056 ± 0.0018	0.0313 ± 0.0296	0.0009 ± 0.0038
SpokenArabicDigits	ACC	0.4902 ± 0.0258	0.1000 ± 0.0000	0.2415 ± 0.0313	0.3869 ± 0.0543	0.4655 ± 0.1073	0.1889 ± 0.0452	0.3692 ± 0.0053	0.1000 ± 0.0000	0.3054 ± 0.0040	0.4420 ± 0.0257
	F1	0.4939 ± 0.0288	0.0182 ± 0.0000	0.2320 ± 0.0343	0.3303 ± 0.0362	0.4917 ± 0.1044	0.1196 ± 0.0519	0.3713 ± 0.0084	0.1819 ± 0.0000	0.3030 ± 0.0060	0.4460 ± 0.0206
	NMI	0.4740 ± 0.0129	0.0000 ± 0.0000	0.1566 ± 0.0474	0.4331 ± 0.0777						

TABLE III
ABLATION STUDIES ON MODULE DESIGNS OF SIGMA, PART I. RED “+” MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (1-15)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
ArticulatoryWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470
			0.7155 ± 0.0137	0.7451 ± 0.0215	0.7824 ± 0.0132	0.5728 ± 0.0248
	✓		0.7739 ± 0.0437	0.7997 ± 0.0481	0.8255 ± 0.0189	0.6570 ± 0.0427
			0.6918 ± 0.0306	0.7336 ± 0.0245	0.7709 ± 0.0187	0.5517 ± 0.0369
			0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492
AtrialFibrillation	✓	✓	0.4867 ± 0.0163	0.4331 ± 0.0116	0.1050 ± 0.0047	0.0404 ± 0.0051
			0.4800 ± 0.0400	0.4681 ± 0.0854	0.1365 ± 0.0448	0.0664 ± 0.0421
	✓		0.4600 ± 0.0327	0.5024 ± 0.0485	0.1000 ± 0.0087	0.0346 ± 0.0071
			0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363
			0.9825 ± 0.0127	0.9824 ± 0.0129	0.9497 ± 0.0366	0.9533 ± 0.0338
BasicMotions	✓	✓	0.8900 ± 0.1317	0.9203 ± 0.0746	0.8924 ± 0.0754	0.8361 ± 0.1360
			0.9700 ± 0.0100	0.9698 ± 0.0102	0.9138 ± 0.0287	0.9202 ± 0.0263
	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576
			0.7824 ± 0.0547	0.8177 ± 0.0436	0.8416 ± 0.0367	0.7078 ± 0.0644
	✓	✓	0.7371 ± 0.0227	0.7610 ± 0.0346	0.8038 ± 0.0198	0.6417 ± 0.0303
Character Trajectories			0.7369 ± 0.0590	0.7699 ± 0.0708	0.8138 ± 0.0468	0.6556 ± 0.0750
	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299
			0.8489 ± 0.0349	0.8739 ± 0.0420	0.8737 ± 0.0444	0.7589 ± 0.0659
	✓	✓	0.7811 ± 0.0658	0.8191 ± 0.0588	0.8346 ± 0.0364	0.6817 ± 0.0702
			0.8078 ± 0.0114	0.8292 ± 0.0241	0.8376 ± 0.0130	0.6937 ± 0.0195
Cricket	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345
			0.3700 ± 0.0179	0.3600 ± 0.0253	0.1143 ± 0.0234	0.0526 ± 0.0108
	✓	✓	0.3280 ± 0.0421	0.2966 ± 0.0425	0.0869 ± 0.0246	0.0196 ± 0.0205
			0.3680 ± 0.0194	0.3449 ± 0.0235	0.1229 ± 0.0145	0.0306 ± 0.0112
	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760
ERing			0.6867 ± 0.0669	0.7380 ± 0.0341	0.6238 ± 0.0426	0.5008 ± 0.0669
	✓	✓	0.7300 ± 0.0485	0.7313 ± 0.0475	0.6198 ± 0.0400	0.5226 ± 0.0569
			0.7173 ± 0.0259	0.7082 ± 0.0253	0.6112 ± 0.0390	0.5033 ± 0.0415
	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726
			0.4131 ± 0.0101	0.3546 ± 0.0347	0.2196 ± 0.0211	0.1907 ± 0.0156
EigenWorms	✓	✓	0.4425 ± 0.0451	0.4077 ± 0.0359	0.1829 ± 0.0256	0.1469 ± 0.0344
			0.3822 ± 0.0462	0.3397 ± 0.0392	0.1713 ± 0.0600	0.1368 ± 0.0716
	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336
			0.4640 ± 0.0291	0.4648 ± 0.0300	0.2217 ± 0.0332	0.1511 ± 0.0259
	✓	✓	0.4720 ± 0.0368	0.4705 ± 0.0382	0.2330 ± 0.0451	0.1551 ± 0.0279
Epilepsy			0.4829 ± 0.0238	0.4847 ± 0.0243	0.2418 ± 0.0298	0.1637 ± 0.0212
	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123
			0.3015 ± 0.0099	0.2777 ± 0.0082	0.0114 ± 0.0033	0.0049 ± 0.0030
	✓	✓	0.3088 ± 0.0183	0.2881 ± 0.0252	0.0184 ± 0.0066	0.0104 ± 0.0062
			0.2950 ± 0.0092	0.2797 ± 0.0108	0.0124 ± 0.0061	0.0045 ± 0.0039
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004
			0.5051 ± 0.0028	0.5050 ± 0.0028	0.0001 ± 0.0001	0.0000 ± 0.0001
	✓	✓	0.5101 ± 0.0023	0.5092 ± 0.0027	0.0003 ± 0.0001	0.0003 ± 0.0002
			0.5063 ± 0.0076	0.5061 ± 0.0077	0.0003 ± 0.0005	0.0003 ± 0.0007
	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010
FingerMovements			0.5168 ± 0.0101	0.5123 ± 0.0056	0.0013 ± 0.0015	0.0008 ± 0.0017
	✓	✓	0.5168 ± 0.0128	0.5125 ± 0.0160	0.0013 ± 0.0014	0.0005 ± 0.0019
			0.5130 ± 0.0074	0.5127 ± 0.0073	0.0006 ± 0.0007	0.0015 ± 0.0010
	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052
			0.3051 ± 0.0154	0.3027 ± 0.0162	0.0139 ± 0.0036	0.0008 ± 0.0029
HandMovement Direction	✓	✓	0.3179 ± 0.0249	0.3148 ± 0.0246	0.0201 ± 0.0119	0.0049 ± 0.0111
			0.3060 ± 0.0126	0.3034 ± 0.0116	0.0144 ± 0.0029	0.0011 ± 0.0025
	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180
			0.2102 ± 0.0132	0.2103 ± 0.0137	0.3030 ± 0.0188	0.0770 ± 0.0111
	✓	✓	0.2444 ± 0.0113	0.2460 ± 0.0156	0.3503 ± 0.0160	0.1050 ± 0.0109
Handwriting			0.2224 ± 0.0130	0.2234 ± 0.0113	0.3212 ± 0.0209	0.0852 ± 0.0100
	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404
			0.6826 ± 0.0369	0.6002 ± 0.0271	0.0438 ± 0.0108	0.0952 ± 0.0277
	✓	✓	0.6841 ± 0.0410	0.5474 ± 0.0747	0.0328 ± 0.0106	0.0620 ± 0.0591
			0.7066 ± 0.0091	0.6463 ± 0.0947	0.0365 ± 0.0108	0.0846 ± 0.0516

clustering. Temporal restoration and prototype-guided refinement are both required after source-level pruning.

The complete ablations to better support the role assignment summarized in the main paper. CCMA restricts which patches participate in early feature interaction, MEVG organizes the admitted content into complementary endogenous views, reconstruction keeps the pruned representation semantically traceable, and contrastive learning links the organized views to cluster geometry. SIGMA therefore does not rely on a single dominant block. Its empirical gains come from a compact chain in which admission, view organization, semantic restoration, and cluster refinement address different failure modes of redundancy-dominated MTS clustering. This role assignment helps disentangle redundancy suppression from semantic preservation, then connects retained semantics to cluster refinement under the prune-and-preserve framework.

C. Additional Training-Loss Curves

The main paper evaluates training stability on four representative datasets. This supplemental material subsection extends

TABLE IV
ABLATION STUDIES ON MODULE DESIGNS OF SIGMA, PART II. RED “+” MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (16-30)	Components		Metrics			
	CCMA	MEVG	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
		✓	0.1351 ± 0.0015	0.1206 ± 0.0036	0.0147 ± 0.0008	0.0064 ± 0.0001
	✓		0.1304 ± 0.0032	0.1115 ± 0.0076	0.0132 ± 0.0011	0.0048 ± 0.0013
			0.1323 ± 0.0039	0.1099 ± 0.0069	0.0139 ± 0.0017	0.0050 ± 0.0011
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
		✓	0.8162 ± 0.0393	0.8557 ± 0.0416	0.8044 ± 0.0126	0.7321 ± 0.0429
	✓		0.8750 ± 0.0347	0.8840 ± 0.0294	0.8220 ± 0.0239	0.7847 ± 0.0385
			0.8800 ± 0.0306	0.8802 ± 0.0270	0.8140 ± 0.0272	0.7648 ± 0.0553
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
		✓	0.2018 ± 0.0106	0.1853 ± 0.0156	0.1475 ± 0.0125	0.0561 ± 0.0074
	✓		0.2096 ± 0.0131	0.1792 ± 0.0080	0.1458 ± 0.0102	0.0567 ± 0.0093
			0.2048 ± 0.0134	0.1824 ± 0.0031	0.1553 ± 0.0062	0.0589 ± 0.0052
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
		✓	0.2839 ± 0.0032	0.2979 ± 0.0079	0.3439 ± 0.0107	0.1012 ± 0.0057
	✓		0.3272 ± 0.0278	0.3460 ± 0.0497	0.4212 ± 0.0300	0.1600 ± 0.0271
			0.2883 ± 0.0069	0.2959 ± 0.0148	0.3481 ± 0.0125	0.1084 ± 0.0080
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
		✓	0.5360 ± 0.0178	0.5196 ± 0.0260	0.0053 ± 0.0041	0.0041 ± 0.0050
	✓		0.5307 ± 0.0224	0.5249 ± 0.0284	0.0042 ± 0.0048	0.0032 ± 0.0066
			0.5238 ± 0.0115	0.4965 ± 0.0034	0.0035 ± 0.0038	0.0007 ± 0.0027
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
		✓	0.4533 ± 0.0457	0.4550 ± 0.0494	0.4481 ± 0.0457	0.2797 ± 0.0460
	✓		0.4528 ± 0.0511	0.4554 ± 0.0583	0.4444 ± 0.0859	0.2844 ± 0.0753
			0.4750 ± 0.0072	0.4797 ± 0.0059	0.4753 ± 0.0118	0.3082 ± 0.0133
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
		✓	0.2768 ± 0.0192	0.2752 ± 0.0175	0.2469 ± 0.0372	0.1170 ± 0.0271
	✓		0.4159 ± 0.0708	0.4722 ± 0.0819	0.4329 ± 0.0928	0.2465 ± 0.0908
			0.2550 ± 0.0164	0.2553 ± 0.0131	0.2111 ± 0.0293	0.0900 ± 0.0244
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
		✓	0.5957 ± 0.0208	0.5949 ± 0.0207	0.5736 ± 0.0164	0.4065 ± 0.0264
	✓		0.6614 ± 0.0378	0.6457 ± 0.0436	0.6188 ± 0.0319	0.4953 ± 0.0431
			0.5954 ± 0.0262	0.5955 ± 0.0250	0.5702 ± 0.0197	0.4034 ± 0.0284
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
		✓	0.1036 ± 0.0047	0.1044 ± 0.0044	0.1576 ± 0.0107	0.0727 ± 0.0029
	✓		0.1036 ± 0.0016	0.1027 ± 0.0021	0.1568 ± 0.0039	0.0289 ± 0.0014
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
		✓	0.6330 ± 0.0454	0.6263 ± 0.0497	0.4037 ± 0.0637	0.3443 ± 0.0751
	✓		0.5340 ± 0.0979	0.5410 ± 0.0966	0.3100 ± 0.1019	0.2449 ± 0.1091
			0.6026 ± 0.0949	0.6012 ± 0.0964	0.3912 ± 0.1121	0.3290 ± 0.1141
SelfRegulation SCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
		✓	0.7647 ± 0.0254	0.7635 ± 0.0258	0.2224 ± 0.0410	0.2816 ± 0.0510
	✓		0.7740 ± 0.0170	0.7730 ± 0.0160	0.2387 ± 0.0412	0.3002 ± 0.0376
			0.7647 ± 0.0103	0.7644 ± 0.0099	0.2150 ± 0.0213	0.2794 ± 0.0224
SelfRegulation SCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
		✓	0.5279 ± 0.0107	0.5270 ± 0.0093	0.0026 ± 0.0022	0.0010 ± 0.0029
	✓		0.5147 ± 0.0106	0.5141 ± 0.0105	0.0010 ± 0.0013	0.0013 ± 0.0017
			0.5142 ± 0.0032	0.5141 ± 0.0033	0.0006 ± 0.0003	0.0018 ± 0.0004
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
		✓	0.4712 ± 0.0196	0.4778 ± 0.0219	0.4407 ± 0.0153	0.2844 ± 0.0124
	✓		0.4477 ± 0.0191	0.4502 ± 0.0231	0.4457 ± 0.0383	0.2802 ± 0.0395
			0.4637 ± 0.0203	0.4691 ± 0.0204	0.4426 ± 0.0119	0.2821 ± 0.0086
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
		✓	0.4000 ± 0.0277	0.3749 ± 0.0372	0.0458 ± 0.0276	0.0173 ± 0.0277
	✓		0.4222 ± 0.0502	0.3944 ± 0.0506	0.0417 ± 0.0306	0.0215 ± 0.0349
			0.4148 ± 0.0544	0.3852 ± 0.0626	0.0547 ± 0.0419	0.0341 ± 0.0161
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
		✓	0.3373 ± 0.0086	0.3156 ± 0.0090	0.1918 ± 0.0200	0.0954 ± 0.0114
	✓		0.3945 ± 0.0411	0.3881 ± 0.0581	0.2768 ± 0.0378	0.1487 ± 0.0261
			0.3127 ± 0.0232	0.2967 ± 0.0171	0.1782 ± 0.0133	0.0860 ± 0.0156

TABLE V
ABLATION STUDIES ON ROUTE DESIGNS OF SIGMA, PART I. RED “+”
MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (1-15)	Components		Metrics				
	Recon	Contra	ACC	F1	NMI	ARI	
ArticulatoryWord Recognition	✓	✓	0.7329 ± 0.0395	0.7696 ± 0.0443	0.8037 ± 0.0246	0.6029 ± 0.0470	
	✓	✓	0.6035 ± 0.0317	0.6212 ± 0.0261	0.7016 ± 0.0278	0.4614 ± 0.0399	
	✓	✓	0.8247 ± 0.0562	0.8514 ± 0.0558	0.8747 ± 0.0426	0.7445 ± 0.0793	
	✓	✓	0.1419 ± 0.0485	0.2366 ± 0.0323	0.2485 ± 0.0639	0.0353 ± 0.0299	
AtrialFibrillation	✓	✓	0.4933 ± 0.0327	0.4720 ± 0.0232	0.1215 ± 0.0474	0.0443 ± 0.0492	
	✓	✓	0.4933 ± 0.0249	0.4763 ± 0.0291	0.0833 ± 0.0242	0.0169 ± 0.0203	
	✓	✓	0.4733 ± 0.0389	0.4520 ± 0.0542	0.0765 ± 0.0312	0.0190 ± 0.0333	
	✓	✓	0.4600 ± 0.0389	0.4518 ± 0.0598	0.0937 ± 0.0356	0.0212 ± 0.0325	
BasicMotions	✓	✓	0.9675 ± 0.0150	0.9671 ± 0.0154	0.9273 ± 0.0277	0.9169 ± 0.0363	
	✓	✓	0.9050 ± 0.1226	0.9319 ± 0.0784	0.9161 ± 0.0911	0.8633 ± 0.1575	
	✓	✓	0.6600 ± 0.0366	0.8105 ± 0.0355	0.7754 ± 0.0499	0.6063 ± 0.0482	
	✓	✓	0.5100 ± 0.0289	0.6355 ± 0.0519	0.5518 ± 0.0478	0.3927 ± 0.0402	
Character Trajectories	✓	✓	0.7879 ± 0.0499	0.8091 ± 0.0408	0.8463 ± 0.0284	0.7214 ± 0.0576	
	✓	✓	0.6714 ± 0.0832	0.7199 ± 0.0942	0.7443 ± 0.0645	0.5830 ± 0.0923	
	✓	✓	0.7389 ± 0.0708	0.7674 ± 0.0674	0.8189 ± 0.0503	0.6750 ± 0.0821	
	✓	✓	0.0651 ± 0.0000	0.1222 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
Cricket	✓	✓	0.8244 ± 0.0255	0.8463 ± 0.0115	0.8522 ± 0.0167	0.7259 ± 0.0299	
	✓	✓	0.7900 ± 0.1095	0.8001 ± 0.1052	0.8279 ± 0.0699	0.6912 ± 0.1158	
	✓	✓	0.7511 ± 0.0708	0.7828 ± 0.0339	0.7682 ± 0.0476	0.5964 ± 0.0836	
	✓	✓	0.1000 ± 0.0333	0.2018 ± 0.0960	0.0721 ± 0.1442	0.0121 ± 0.0241	
DuckDuckGeese	✓	✓	0.3900 ± 0.0141	0.3730 ± 0.0193	0.1749 ± 0.0375	0.0795 ± 0.0345	
	✓	✓	0.4740 ± 0.0546	0.4690 ± 0.0553	0.2439 ± 0.0457	0.1618 ± 0.0523	
	✓	✓	0.3880 ± 0.0306	0.3682 ± 0.0377	0.1562 ± 0.0375	0.0621 ± 0.0248	
	✓	✓	0.2000 ± 0.0000	0.3333 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
ERing	✓	✓	0.7307 ± 0.0629	0.7287 ± 0.0680	0.5996 ± 0.0590	0.5124 ± 0.0760	
	✓	✓	0.7533 ± 0.0419	0.7553 ± 0.0421	0.6735 ± 0.0240	0.5854 ± 0.0353	
	✓	✓	0.8020 ± 0.0937	0.8006 ± 0.0951	0.7408 ± 0.0702	0.6688 ± 0.1070	
	✓	✓	0.1667 ± 0.0000	0.2857 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
EigenWorms	✓	✓	0.4193 ± 0.0788	0.3646 ± 0.0438	0.1612 ± 0.0594	0.1515 ± 0.0726	
	✓	✓	0.4556 ± 0.0523	0.4150 ± 0.0632	0.2774 ± 0.0369	0.2479 ± 0.0381	
	✓	✓	0.3382 ± 0.0317	0.3009 ± 0.0439	0.0843 ± 0.0295	0.0512 ± 0.0158	
	✓	✓	0.4247 ± 0.0000	0.5962 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
Epilepsy	✓	✓	0.4655 ± 0.0229	0.4573 ± 0.0233	0.2355 ± 0.0362	0.1668 ± 0.0336	
	✓	✓	0.5716 ± 0.0429	0.5746 ± 0.0356	0.3998 ± 0.0509	0.3182 ± 0.0499	
	✓	✓	0.4320 ± 0.0292	0.4259 ± 0.0276	0.2047 ± 0.0333	0.1442 ± 0.0279	
	✓	✓	0.4291 ± 0.0332	0.4217 ± 0.0355	0.1670 ± 0.0517	0.1224 ± 0.0370	
Ethanol Concentration	✓	✓	0.3061 ± 0.0208	0.2766 ± 0.0066	0.0199 ± 0.0156	0.0108 ± 0.0123	
	✓	✓	0.2916 ± 0.0080	0.2841 ± 0.0078	0.0129 ± 0.0041	0.0032 ± 0.0014	
	✓	✓	0.3116 ± 0.0143	0.2799 ± 0.0252	0.0308 ± 0.0187	0.0144 ± 0.0106	
	✓	✓	0.3179 ± 0.0130	0.3277 ± 0.0507	0.0442 ± 0.0192	0.0238 ± 0.0127	
FaceDetection	✓	✓	0.5107 ± 0.0042	0.5100 ± 0.0049	0.0004 ± 0.0003	0.0004 ± 0.0004	
	✓	✓	0.5090 ± 0.0055	0.5086 ± 0.0056	0.0003 ± 0.0003	0.0003 ± 0.0004	
	✓	✓	0.5000 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
	✓	✓	0.5000 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
FingerMovements	✓	✓	0.5188 ± 0.0085	0.5142 ± 0.0086	0.0013 ± 0.0007	0.0006 ± 0.0010	
	✓	✓	0.5135 ± 0.0097	0.5114 ± 0.0111	0.0008 ± 0.0009	0.0013 ± 0.0012	
	✓	✓	0.5231 ± 0.0102	0.5188 ± 0.0102	0.0019 ± 0.0012	0.0002 ± 0.0016	
	✓	✓	0.5000 ± 0.0000	0.6667 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	
HandMovement Direction	✓	✓	0.3333 ± 0.0143	0.3298 ± 0.0171	0.0205 ± 0.0063	0.0060 ± 0.0052	
	✓	✓	0.3077 ± 0.0162	0.3061 ± 0.0171	0.0148 ± 0.0062	0.0005 ± 0.0053	
	✓	✓	0.3111 ± 0.0208	0.3077 ± 0.0221	0.0215 ± 0.0075	0.0053 ± 0.0072	
	✓	✓	0.3068 ± 0.0068	0.2901 ± 0.0134	0.0201 ± 0.0043	0.0021 ± 0.0020	
Handwriting	✓	✓	0.2386 ± 0.0181	0.2433 ± 0.0195	0.3461 ± 0.0286	0.1006 ± 0.0180	
	✓	✓	0.1706 ± 0.0061	0.1741 ± 0.0083	0.2227 ± 0.0045	0.0385 ± 0.0019	
	✓	✓	0.2054 ± 0.0071	0.2071 ± 0.0059	0.2809 ± 0.0091	0.0663 ± 0.0054	
	✓	✓	0.1428 ± 0.0060	0.1408 ± 0.0057	0.1813 ± 0.0062	0.0217 ± 0.0024	
Heartbeat	✓	✓	0.6636 ± 0.0589	0.5679 ± 0.0507	0.0391 ± 0.0232	0.0689 ± 0.0404	
	✓	✓	0.5628 ± 0.0376	0.5053 ± 0.0802	0.0459 ± 0.0155	0.0065 ± 0.0412	
	✓	✓	0.7002 ± 0.0176	0.7537 ± 0.1522	0.0170 ± 0.0068	0.0161 ± 0.0034	
	✓	✓	0.7213 ± 0.0000	0.8381 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	

TABLE VI
ABLATION STUDIES ON ROUTE DESIGNS OF SIGMA, PART II. RED “+”
MARKS DEGRADATION FROM COMPLETE SIGMA.

Datasets (16-30)	Components		Metrics			
	Recon	Contra	ACC	F1	NMI	ARI
InsectWingbeat	✓	✓	0.1344 ± 0.0023	0.1202 ± 0.0033	0.0153 ± 0.0012	0.0061 ± 0.0007
	✓	✓	0.1206 ± 0.0063	0.1104 ± 0.0111	0.0053 ± 0.0027	0.0025 ± 0.0014
	✓	✓	0.1000 ± 0.0000	0.1818 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1000 ± 0.0000	0.1818 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
JapaneseVowels	✓	✓	0.8647 ± 0.0447	0.8626 ± 0.0432	0.8195 ± 0.0260	0.7682 ± 0.0565
	✓	✓	0.8859 ± 0.0310	0.8942 ± 0.0270	0.8224 ± 0.0315	0.7627 ± 0.0522
	✓	✓	0.6441 ± 0.1450	0.6632 ± 0.1467	0.5833 ± 0.1473	0.4328 ± 0.1564
	✓	✓	0.1844 ± 0.0000	0.3113 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
LSST	✓	✓	0.2042 ± 0.0087	0.1805 ± 0.0063	0.1456 ± 0.0063	0.0555 ± 0.0044
	✓	✓	0.3362 ± 0.0017	0.2228 ± 0.0094	0.1699 ± 0.0142	0.0714 ± 0.0062
	✓	✓	0.1969 ± 0.0139	0.1727 ± 0.0102	0.1427 ± 0.0084	0.0537 ± 0.0075
	✓	✓	0.3155 ± 0.0000	0.4797 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
Libras	✓	✓	0.3156 ± 0.0374	0.3162 ± 0.0405	0.4101 ± 0.0302	0.1502 ± 0.0253
	✓	✓	0.3600 ± 0.0283	0.3572 ± 0.0231	0.4518 ± 0.0285	0.1916 ± 0.0259
	✓	✓	0.4161 ± 0.0210	0.4303 ± 0.0315	0.5402 ± 0.0261	0.2622 ± 0.0231
	✓	✓	0.2644 ± 0.0235	0.2673 ± 0.0222	0.3222 ± 0.0456	0.0927 ± 0.0307
MotorImagery	✓	✓	0.5323 ± 0.0147	0.5271 ± 0.0154	0.0039 ± 0.0034	0.0025 ± 0.0046
	✓	✓	0.5079 ± 0.0044	0.4726 ± 0.0036	0.0004 ± 0.0004	0.0016 ± 0.0004
	✓	✓	0.5270 ± 0.0074	0.5048 ± 0.0179	0.0034 ± 0.0023	0.0009 ± 0.0017
	✓	✓	OOM	OOM	OOM	OOM
NATOPS	✓	✓	0.4756 ± 0.0180	0.4638 ± 0.0129	0.4697 ± 0.0383	0.3103 ± 0.0385
	✓	✓	0.4972 ± 0.0478	0.4940 ± 0.0434	0.4796 ± 0.0594	0.3245 ± 0.0508
	✓	✓	0.4328 ± 0.0428	0.4242 ± 0.0431	0.3578 ± 0.0955	0.2377 ± 0.0650
	✓	✓	0.1967 ± 0.0600	0.3526 ± 0.1337	0.0770 ± 0.1539	0.0221 ± 0.0441
PEMS-SF	✓	✓	0.3905 ± 0.0724	0.4364 ± 0.0969	0.3625 ± 0.1185	0.2028 ± 0.0839
	✓	✓	0.2368 ± 0.0216	0.2476 ± 0.0355	0.1705 ± 0.0511	0.0609 ± 0.0218
	✓	✓	0.2700 ± 0.0273	0.2785 ± 0.0369	0.1930 ± 0.0781	0.0771 ± 0.0412
	✓	✓	0.1477 ± 0.0000	0.2574 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
PenDigits	✓	✓	0.6796 ± 0.0497	0.6724 ± 0.0488	0.6440 ± 0.0381	0.5249 ± 0.0488
	✓	✓	0.7139 ± 0.0446	0.7030 ± 0.0448	0.6472 ± 0.0277	0.5420 ± 0.0420
	✓	✓	0.1041 ± 0.0000	0.1885 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1041 ± 0.0000	0.1885 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
PhonemeSpectra	✓	✓	0.1125 ± 0.0040	0.1119 ± 0.0039	0.1765 ± 0.0084	0.0333 ± 0.0020
	✓	✓	0.0985 ± 0.0017	0.0967 ± 0.0012	0.1443 ± 0.0051	0.0243 ± 0.0012
	✓	✓	0.0256 ± 0.0000	0.0500 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.0256 ± 0.0000	0.0500 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
RacketSports	✓	✓	0.5908 ± 0.0704	0.5918 ± 0.0700	0.3431 ± 0.0775	0.2754 ± 0.0886
	✓	✓	0.6389 ± 0.0424	0.6436 ± 0.0345	0.4214 ± 0.0229	0.3526 ± 0.0483
	✓	✓	0.4488 ± 0.0172	0.4449 ± 0.0185	0.1929 ± 0.0490	0.1187 ± 0.0352
	✓	✓	0.4244 ± 0.0367	0.3947 ± 0.0725	0.1550 ± 0.0234	0.1061 ± 0.0245
Self RegulationSCP1	✓	✓	0.7747 ± 0.0140	0.7738 ± 0.0136	0.2371 ± 0.0269	0.3014 ± 0.0296
	✓	✓	0.7458 ± 0.0132	0.7421 ± 0.0156	0.2011 ± 0.0274	0.2411 ± 0.0261
	✓	✓	0.7337 ± 0.0714	0.7246 ± 0.0859	0.1956 ± 0.0777	0.2376 ± 0.1041
	✓	✓	0.7480 ± 0.0175	0.7477 ± 0.0176	0.1878 ± 0.0261	0.2458 ± 0.0341
Self RegulationSCP2	✓	✓	0.5253 ± 0.0151	0.5162 ± 0.0079	0.0036 ± 0.0050	0.0010 ± 0.0042
	✓	✓	0.5347 ± 0.0139	0.5325 ± 0.0152	0.0041 ± 0.0021	0.0030 ± 0.0029
	✓	✓	0.5158 ± 0.0173	0.5106 ± 0.0157	0.0017 ± 0.0031	0.0003 ± 0.0040
	✓	✓	0.5200 ± 0.0061	0.5191 ± 0.0059	0.0013 ± 0.0007	0.0009 ± 0.0009
Spoken ArabicDigits	✓	✓	0.4260 ± 0.0202	0.4328 ± 0.0168	0.3868 ± 0.0165	0.2365 ± 0.0194
	✓	✓	0.3631 ± 0.0512	0.3642 ± 0.0468	0.3212 ± 0.0486	0.1523 ± 0.0453
	✓	✓	0.1000 ± 0.0000	0.1819 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
	✓	✓	0.1000 ± 0.0000	0.1819 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000
StandWalkJump	✓	✓	0.4000 ± 0.0637	0.3740 ± 0.0724	0.0441 ± 0.0480	0.0171 ± 0.0522
	✓	✓	0.4593 ± 0.0378	0.4537 ± 0.0361	0.0530 ± 0.0194	0.0194 ± 0.0251
	✓	✓	0.4519 ± 0.0363	0.4355 ± 0.0497	0.0543 ± 0.0236	0.0178 ± 0.0199
	✓	✓	0.4296 ± 0.0181	0.3740 ± 0.0214	0.0924 ± 0.0506	0.0016 ± 0.0243
UWaveGesture Library	✓	✓	0.4164 ± 0.0435	0.4043 ± 0.0445	0.3456 ± 0.0479	0.1996 ± 0.0452
	✓	✓	0.4023 ± 0.0342	0.3992 ± 0.0359	0.2887 ± 0.0371	0.1684 ± 0.0280
	✓	✓	0.6318 ± 0.0456	0.6520 ± 0.0381	0.5779 ± 0.0160	0.4497 ± 0.0166
	✓	✓	0.1250 ± 0.0000	0.2222 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000

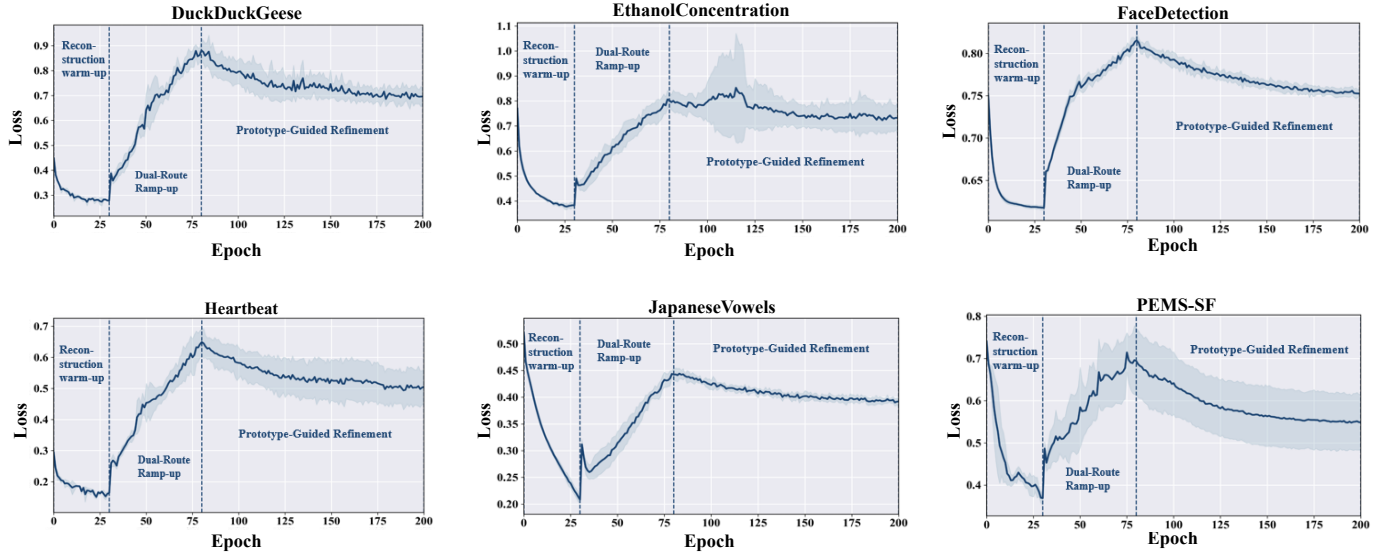


Fig. 2. Additional training-loss curves during SIGMA training on six UEA datasets. Each curve reports the mean loss across five random seeds, and the shaded band denotes standard deviation. Vertical dashed lines mark the semantic warm-up, dual-route ramp-up, and prototype-guided refinement stages.

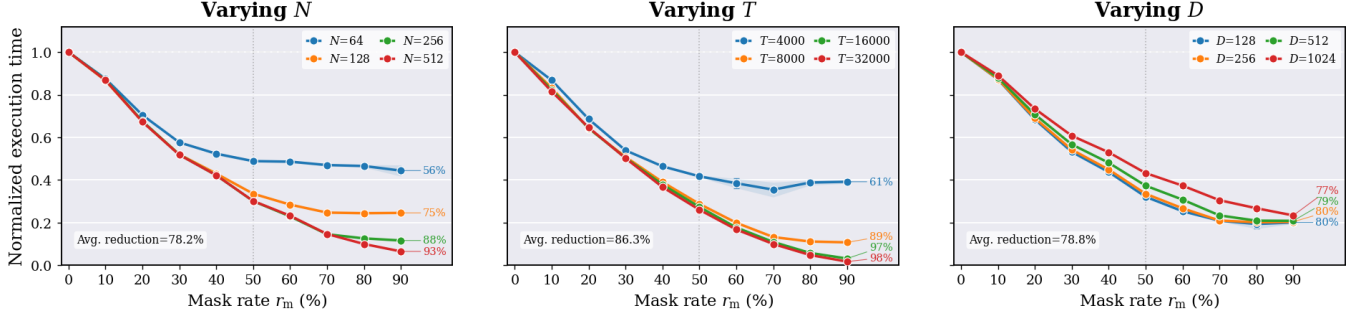


Fig. 3. Scalability under increasing mask rates. The three panels vary N , T , and D , respectively. For each setting, execution time is normalized by the cost at zero mask rate. Shaded bands denote standard deviation, and endpoint annotations report the reduction at 90% mask rate.

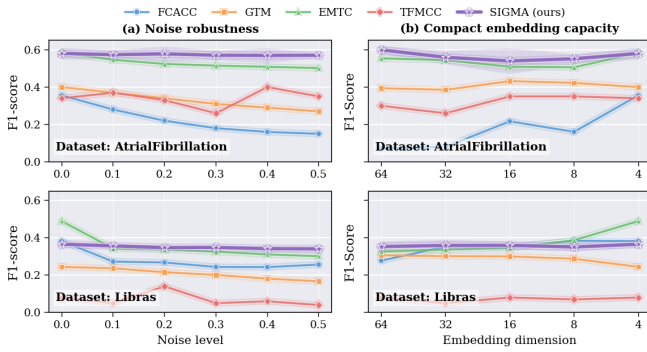


Fig. 4. **Robustness and compactness diagnostics.** F1-score on AtrialFibrillation and Libras under (a) increasing Gaussian noise and (b) reduced embedding dimensions from 64 to 4. SIGMA keeps stable accuracy in both settings, showing that retained representations remain usable under perturbation and compact latent budgets.

missible input set before redundant patches shape feature interaction.

E. Robustness and Compactness Diagnostics

Fig. 4 supplements Q4 by testing whether the retained representation remains usable under input perturbation and compact latent budgets. In Fig. 4(a), Gaussian noise is added to AtrialFibrillation and Libras. SIGMA keeps a smooth F1 trajectory as the noise level increases, especially on AtrialFibrillation, where the curve stays above the compared baselines. On Libras, several methods fluctuate or decline after perturbation, whereas SIGMA remains stable across noise levels. The result suggests that confidence-calibrated admission filters part of the corrupted or routine content before feature interaction, and the retained views still provide sufficient context for clustering.

Fig. 4(b) further reduces the embedding dimension from 64 to 4. SIGMA maintains competitive F1-score under compressed dimensions on both datasets, showing that the learned cluster structure does not rely on an oversized latent space. EMTC remains competitive on Libras at the smallest dimension, but SIGMA gives steadier behavior across the full

dimension range. Together, the two diagnostics indicate that the prune-but-preserve design supports perturbation tolerance and compact representation learning, which is essential for efficient MTS organization under constrained computation and storage budgets.

REFERENCES

- [1] C. Wang, M. Du, X. Jiang, and Y. Dong, “Fuzzy cluster-aware contrastive clustering for time series,” *Pattern Recognition*, vol. 173, p. 112899, 2026.
- [2] C. Wang, M. Du, and X. Jiang, “Time-frequency augmented multi-level contrastive clustering for time series,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 31, 2026, pp. 26 142–26 150.
- [3] C. HE, X. Huang, G. Jiang, Z. Li, D. Lian, H. Xie, E. Chen, xijie liang, Zhengzengrong, and P. Lee, “GTM: A general time-series model for enhanced representation learning of time-series data,” in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2026. [Online]. Available: <https://openreview.net/forum?id=PWM6FERWz9>
- [4] Z. Tan, X. Luo, Y. Liu, and Y. Zhang, “Mask the redundancy: Evolving masking representation learning for multivariate time-series clustering,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 40, no. 30, 2026, pp. 25 787–25 795.
- [5] E. Fu and Y. Hu, “Frequency-masked embedding inference: A non-contrastive approach for time series representation learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 39, no. 16, 2025, pp. 16 639–16 647.
- [6] Y. Li, M. Du, X. Jiang, and N. Zhang, “Contrastive learning-based multi-view clustering for incomplete multivariate time series,” *Information Fusion*, vol. 117, p. 102812, 2025.
- [7] P. Boniol, D. Tiano, A. Bonifati, and T. Palpanas, “ k -graph: A graph embedding for interpretable time series clustering,” *IEEE Transactions on Knowledge and Data Engineering*, vol. 37, no. 5, pp. 2680–2694, 2025.
- [8] J. Liu and S. Chen, “Timesurl: Self-supervised contrastive learning for universal time series representation learning,” in *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, vol. 38, no. 12, 2024, pp. 13 918–13 926.
- [9] S. Gao, T. Koker, O. Queen, T. Hartvigsen, T. Tsiligkaridis, and M. Zitnik, “Units: A unified multi-task time series model,” in *Proceedings of the Annual Conference on Neural Information Processing Systems (NeurIPS)*, vol. 37, 2024, pp. 140 589–140 631.